



Origins of extreme boundary lubrication by phosphatidylcholine liposomes



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ABSTRACT

Phosphatidylcholine (PC) vesicles have been shown to have remarkable boundary lubricating properties under physiologically-high pressures. Here we carry out a systematic study, using a surface force balance, of the normal and shear (frictional) forces between two opposing surfaces bearing different PC vesicles across water, to elucidate the origin of these properties. Small unilamellar vesicles (SUVs, diameters < 100 nm) of the symmetric saturated diacyl PCs DMPC (C₁₄), DPPC (C₁₆) and DSPC (C₁₈) attached to mica surfaces were studied in their solid-ordered (SO) phase on the surface. Overall liposome lubrication ability improves markedly with increasing acyl chain length, and correlates strongly with the liposomes' structural integrity on the substrate surface: DSPC-SUVs were stable on the surface, and provided extremely efficient lubrication (friction coefficient $\mu \approx 10^{-4}$) at room temperature at pressures up to at least 18 MPa. DMPC-SUVs ruptured following adsorption, providing poor high-pressure lubrication, while DPPC-SUVs behavior was intermediate between the two. These results can be well understood in terms of the hydration-lubrication paradigm, but suggest that an earlier conjecture, that highly-efficient lubrication by PC-SUVs depended simply on their being in the SO rather than in the liquid-disordered phase, should be more nuanced. Our results indicate that the resistance of the SUVs to mechanical deformation and rupture is the dominant factor in determining their overall boundary lubrication efficiency in our system.

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1. Introduction

Liposomes are closed, largely-spherical vesicles of colloidal dimensions, consisting of phospholipids (PL) that form a self-closed bilayer membrane. Due to their composition and structure they are biocompatible, and thus serve as a useful tool for biomedical applications including drug delivery systems, signal enhancers in medical diagnostics, vaccines, gene delivery systems and solubilizers for different substances [1–3]. Liposomes and phospholipid bilayers are also an important model system in basic studies of biological membranes [4], as phospholipids are the dominant components of such bilayer membranes and are responsible for many of their unique physicochemical and biochemical properties [5,6]. PLs have been conjectured to lubricate articulating cartilage [7–10] in the classical boundary lubrication mode, where the surfactant alkyl tails are exposed at each surface and slide past each

other [7]. Recently, however [11], it was shown that, for boundary lubrication between self-assembled layers of polar surfactants under water, the slip-plane shifted from the mid-plane between the surfactant layers to the headgroup/substrate interface. This is in contrast to classical boundary lubrication [12,13] (which was assumed also for PLs on cartilage [7–10]), and resulted from the hydration of the polar-head-groups, and the consequent hydration lubrication acting at that interface [11,14].

Several studies of lubrication by PL layers or multilayers, and by surface-attached liposomes have since been reported [15–19]. Low friction was demonstrated with 1,2-dimyristoyl-*sn*-glycero-3-phosphocholine (DMPC) bilayers using lateral force microscopy [15]. In another study, friction properties of liquid phase phosphatidylcholine (PC) bilayers were examined by means of a tribometer. A friction coefficient of $\mu = 0.035$ was demonstrated in Tris buffer and the importance of mechanical stability under shear or normal load for good lubrication performance was emphasized [16]. The importance of lubricant stability was emphasized in another work where 1,2-dioleoyl-*sn*-glycero-3-phosphatidylcholine (DOPC) bilayers were studied. In this work it was also demonstrated that liquid phase bilayers are less efficient lubricants than gel phase bilayers [17].

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Friction and wear tests using cartilage on cartilage configuration with lubricating fluids containing vesicles composed of different phospholipids or their mixtures (both small unilamellar vesicles, SUVs, and multilamellar vesicles, MLVs) showed static and dynamic friction coefficients (at 37 °C) down to around 0.02 and 0.01, respectively [18]. Lower values of $\mu = 0.002$ were measured on soft (hydrogel) surfaces with adsorbed 1,2-dipalmitoyl-*sn*-glycero-3-phosphatidylcholine (DPPC) lipid bilayers in the gel phase at mean contact pressures of 0.3 MPa [19], while in the same study layers of DOPC (in their liquid phase) were shown to be poorer boundary lubricants (in Ref. [19] a review of several earlier friction studies with PL is given). This was attributed to the DPPC bilayers' better resistance due to being in the gel (or solid-ordered, SO) phase (in contrast to DOPC which was in its liquid disordered, LD phase) and to the hydration lubrication effect [20,21], that will be later discussed in more detail. A common feature of all these earlier studies [15–19] between opposing PL layers or multilayers was the low maximal mean pressure attained between the sliding surfaces, with none – as far as we can ascertain – exceeding about 1–2 MPa. This is notable because pressures in the major joints, where PLs have been conjectured to play a lubricating role [7,8,22], attain values of ca. 5 MPa in normal conditions, with maximal pressures in hips recorded up to ca. 18 MPa [23].

Very recently, it was discovered that SUVs of hydrogenated soy phosphatidylcholine (HSPC), attached as close-packed layers on solid (mica) substrates sliding past each other across aqueous media, could provide extremely efficient lubrication at physiologically-high pressures [24,25]. Friction coefficients as low as $\mu \approx 2 \times 10^{-4}$ – 2×10^{-5} were measured at pressures of up to at least 12 MPa, and attributed to the hydration lubrication mechanism arising from the highly-hydrated phosphocholine headgroups exposed at the outer surfaces of the HSPC vesicles. The persistence of the efficient lubrication up to high pressures was attributed to the smoothness and uniformity of the close-packed layers of the adsorbed vesicles, and in particular to the robustness to pressure of the liposomes. Much higher friction coefficients ($\mu = 0.065$) were measured for SUVs of 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphocholine (POPC) liposomes [25], while at the same time the POPC vesicles showed evidence of rupture and squeeze-out from between the surfaces. HSPC (for which the SO-to-LD transition temperature for *unsupported*, or free-standing bilayers (FSBs), $T_M(\text{HSPC}) = 53$ °C) is in its SO phase at the room temperature of the studies, while POPC ($T_M(\text{POPC}) = -3$ °C) was in the LD phase (we discuss later the important issue of SO-to-LD phase transitions of *supported* bilayers, which is the relevant configuration in our study). It was conjectured, therefore [25], that surface-adsorbed layers of PC-SUVs would be highly-efficient lubricants up to high pressures when in the SO phase, but would be poor lubricants, particularly at higher pressures, if they were in their LD phase. However, whether this conjecture is correct, and indeed the entire issue of how the structure of PC phospholipids, and their corresponding vesicle structure and phase state, is related to efficient lubrication by surface-attached liposomes at high pressures, remains unknown. In the present study we endeavor to gain further insight into the origin of high-pressure lubrication by liposomes through a systematic study of a series of PC-SUVs, using a Surface Force Balance (SFB) in parallel with determination of their microscopic structure on the surface. We examine the lubricating behavior of surface-attached liposomes composed of three PC lipids with increasing acyl chain length: DMPC ([C₁₄]₂), DPPC ([C₁₆]₂) and DSPC ([C₁₈]₂). DPPC and DSPC are examined in the gel phase, and DMPC is studied in the SO phase and also in the mixed phase within the range of the SO-to-LD transition. Together with parallel examination of the structure of the surface-attached vesicles using atomic force microscopy (AFM) and cryo-scanning electron

microscopy (cryo-SEM) we are able to shed strong light on the origins and determinants of friction reduction by surface-adsorbed PC vesicles.

2. Experimental techniques

2.1. Liposome preparation

The lipids, 1,2-dimyristoyl-*sn*-glycero-3-phosphatidylcholine (DMPC, 14:0), 1,2-dipalmitoyl-*sn*-glycero-3-phosphatidylcholine (DPPC 16:0) and 1,2-distearoyl-*sn*-glycero-3-phosphatidylcholine (DSPC, 18:0), were purchased from Lipoid (Ludwigshafen, Germany). The main SO-to-LD transition temperatures of unsupported bilayers of these PCs are 24 °C, 41 °C and 54 °C respectively [26]. SUVs of each PC were prepared using standard approaches [25,27], both in our lab and by Y. Barenholz, R. Cohen and H. Kanaan in the Biochemistry Department of the Hadassah Medical School of the Hebrew University, Jerusalem. Briefly, lipids were dispersed in water, bath-sonicated for 5 min and homogenized for 5 min at the appropriate temperature above the main phase transition of each lipid in order to obtain dispersed multilamellar vesicles (MLV). Next, the MLVs were progressively downsized using an extruder (Northern lipid Inc, Burnaby, BC, Canada) through polycarbonate filters having defined pore sizes starting with 400 nm (3 cycles), 100 nm (4 cycles) and ending with 50 nm (10 cycles). The PC-SUVs were characterized by dynamic light scattering (DLS) and subsequent to their adsorption on the surface, also by AFM and cryo-SEM. The DLS-derived dimensions of the solution-dispersed liposomes were in line with those estimated from AFM and cryo-SEM micrographs of the surface-attached vesicles (see later), though these latter had a spread of dimensions on the surfaces due to the adsorption process [25]. The water used throughout both for the liposome preparation and subsequent measurements was highly-purified (so-called conductivity) water from a Barnstead NanoPure system, with total organic content (TOC) ca. 1 ppb and resistivity 18.2 MΩ.

2.2. Preparation of liposome-coated surfaces

Liposome covered mica surfaces were prepared as follows. Freshly cleaved mica (mounted on cylindrical lenses for use in the SFB, see below) was placed in a 0.3 mm SUV liposome dispersion prepared with Barnstead purified conductivity water. Incubation was at room temperature, except for DMPC which was incubated at 20 °C (and at 28 °C in one experiment, as described below). After overnight incubation, the surfaces were rinsed to remove excess material by placing them in a beaker containing 300 ml water for 30 min, along with a delicate shaking motion. Surfaces prepared for SFB always passed the air–water interface, whereas for examination in the AFM, the surfaces did not pass an interface unless the effect of passing an interface was examined as described later. It is relevant for later understanding to note that, on withdrawal from the dispersion and subsequently from the rinsing beaker, the surfaces were partly coated by water in the form of wet patches separated by dry regions.

2.3. Atomic force microscopy (AFM)

Imaging of surfaces was carried out with an MFP-3D SA (AFM) instrument (Asylum Research, Santa Barbara, CA). Scanning in tapping mode in conductivity water was done using a silicon nitride V-shaped 115 μm long cantilever having a nominal spring constant of 0.35 N/m with a pyramidal silicon nitride tip (NP, Bruker).

2.4. Cryo SEM

Samples were plunged into liquid ethane using a custom made spring-loaded plunger. The size of the mica surfaces with the adsorbed liposomes was ca. 4 mm × 2 mm; they were generally covered by a water droplet which was gently imbibed by tissue prior to freezing. The frozen samples were mounted on a holder and transferred to a BAF 60 freeze fracture device using a VCT 100 Vacuum Cryo Transfer device (both instruments from Leica Microsystems, Vienna), etched at –80 °C for 2 h and coated with 3 nm Pt/C by double axis rotary shadowing. Samples were transferred to an Ultra 55 SEM (Zeiss, Germany) using the VCT 100 and were observed at a temperature of –120 °C at 2 kV using an in-lens secondary electron detector.

The electron microscopy studies were conducted at the Irving and Cherna Moskowitz Center for Nano and Bio-Nano Imaging at the Weizmann Institute of Science.

2.5. Surface Force Balance (SFB)

The technique and detailed procedures for the measurement of normal (F_n) and shear (F_s) forces between molecularly-smooth mica surfaces, half-silvered on their back side, in a crossed cylinder configuration (mean radius of curvature R), a closest separation D apart have been described in detail previously [28]. D is optimally measured to ± 2 – 3 Å by monitoring the wavelength of optical interference fringes of equal chromatic order (FECO). The results presented here are based on nine separate experiments, each with several different contact points between the interacting

surfaces, carried out in temperature-stabilized rooms ($\pm 0.2^\circ\text{C}$ about the chosen temperature). These included two DMPC experiments at 20°C (and additional measurements at 28°C as described); and four DPPC experiments and two DSPC experiments, at room temperature (25°C).

Shear profiles were taken for the three different PC systems by directly measuring the response of the lower surface to shear (lateral) motion applied to the upper surface. Shear amplitudes Δx_0 in the range 200–1200 nm and shear velocities v_s in the range 10–600 nm/s were applied. Shear profiles were taken simultaneously with normal force profiles by applying shear motion at several separations at progressively increasing normal loads. Shear forces F_s are measured from the plateau regime of the friction-force vs. time traces (shown later), where F_s is independent of the shear amplitude. The magnitude of the weakest shear forces is determined either by filtering the signal about the drive frequency, or by fast Fourier transform of the data to yield F_s at the drive frequency, see e.g. Refs. [20,21]; the two approaches yield similar values of F_s . We emphasize that all experiments in the SFB, as well as AFM scans and cryo-SEM imaging, were carried out with the surfaces immersed in conductivity water (no added salt).

The normal compressive loads applied during the experiments cause an elastic flattening of the curved surfaces, due mostly to compression of the glue supporting the mica sheet. This is clearly observed as flattening of radius a (of up to around $40\ \mu\text{m}$) at the tips of the interference fringes. This pressure has a distribution across the contact area with a maximum at the center [29], while the mean pressure P between the compressed surfaces can be directly evaluated from the dimensions of this flattened area:

$$P = F_n/A = F/\pi a^2 \quad (1)$$

The flattening upon contact for non-adhering surfaces can be also evaluated from Hertz contact mechanics theory [29], if one assumes a sphere on a flat geometry (closely equivalent to the crossed cylinder configuration of the mica surfaces):

$$a^3 = \frac{RF}{K} \quad (2)$$

where a is the contact radius, F_n is the applied load and K is the effective elastic modulus of the mica/glue sandwich, determined separately as $K = (5 \pm 2) \times 10^9\ \text{N/m}^2$. Measurement of the area via the fringe tip flattening was sometimes significantly different from that evaluated from the Hertzian expression (see e.g. comparison of values in the caption to Fig. 7). This could be due to uncertainty in the measured a values, particularly at lower loads where a is small, or to the fact that K may vary between contact points and experiments (depending on the mica sheet and local glue thickness). Using the Hertzian expression to evaluate the pressures, the relative values of P are correct for a given contact point at different loads (e.g. in Figs. 7), but for the reasons above we estimate the absolute magnitudes of P to have an uncertainty of $\pm 30\%$.

In the case that the surfaces approached and were adhesively coupled (for the DMPC-coated surfaces), the adhesion energy was calculated from the pull-off force needed to separate them, using the Johnson-Kendall-Roberts (JKR) theory [29]:

$$F_{\text{pulloff}} = 3\pi R\gamma \quad (3)$$

where R is the radius of the curved uncompressed surfaces and γ is the interfacial energy which is by definition half of the adhesion energy.

3. Results

3.1. Characterization

3.1.1. AFM

Surfaces with adsorbed liposomes made of the three different types of lipids were characterized by AFM and cryo SEM. Very different results were obtained for the different lipids. Fig. 1(A) shows an AFM scan of a mica surface that has been coated with DMPC liposomes. No individual liposomes are seen, indicating that liposome rupture and fusion has occurred spontaneously to form a lipid bilayer. This lamellar phase is continuous with some holes (black in the image) of 2 nm depth, as can be seen in the inset of Fig. 1(A). These holes are most likely one monolayer deep. Similar holes, 2.6 nm deep, were previously observed in DPPE-DLPE bilayers [30] and attributed accordingly.

Fig. 1(B) shows DPPC liposomes closely packed and stably adsorbed to the mica surface. Most of the liposomes (gray), as can be seen from the cross section in the inset, have been flattened upon adsorption to a height of 10–15 nm, from a dispersion diameter of ca. 60 nm. This phenomenon of liposome flattening upon adsorption has been well documented [25,31–33]. In order to examine the effect of passage through the air–water interface on the adsorbed vesicles, the sample shown in Fig. 1(B) was taken out of the water and then re-immersed. The result, as presented in Fig. 1(C), shows that this can result in vesicle fusion and formation of a continuous lamellar phase (though not all the liposome-coated surface may have undergone such rupture, see later). The depth of the holes suggests that this continuous phase is a bilayer of thickness of roughly 4.5 nm, which is in good agreement with the literature [34]. Similar rupture of liposomes has been demonstrated before, where adsorbed DPPC vesicles ruptured as a result of passing air–water interface and blowing on the surface with nitrogen gas [35]. In contrast, DSPC-based liposomes were found to be stable on the surface (and similar to Fig. 1(D)) even when crossing an air–water interface. These differences in liposome robustness are strongly correlated with their lubrication abilities, as will be described. We note that both for the DPPC-SUVs and for the DSPC-SUVs the measured heights may also reflect compression by the tip. This is also in line with similar observations on AFM-imaged HSPC-SUVs [25].

3.1.2. Cryo SEM

Fig. 2 shows cryo-SEM images of DPPC and DSPC liposomes. DMPC liposomes could not be imaged with this technique, probably

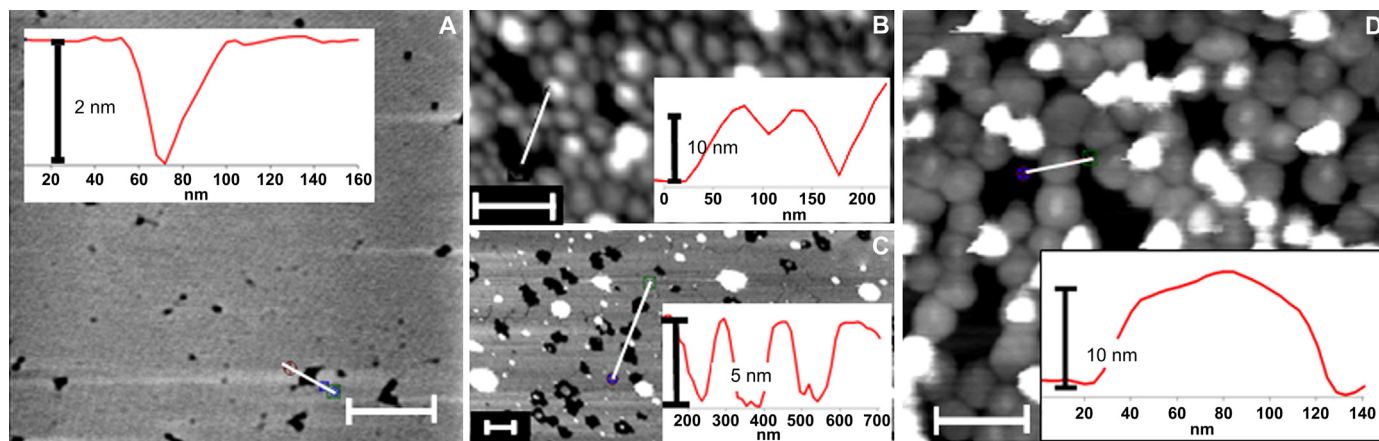


Fig. 1. AFM images of PC liposomes after adsorption to mica, all scans done under water. All scale bars are 200 nm. Insets are cross sections taken at locations marked with white lines in the images. (A) AFM image of DMPC liposomes. The liposomes rupture upon adsorption to form a continuous lamellar phase. Inset: cross section of a hole, depth 2 nm, as shown in the image. (B) Intact DPPC liposomes, with section shown. White spots are excess material adsorbed on top. (C) DPPC after passing air–water interface. A continuous phase is formed, with holes of depth ca. 5 nm. (D) Image of DSPC liposomes, with height of order 10 nm.

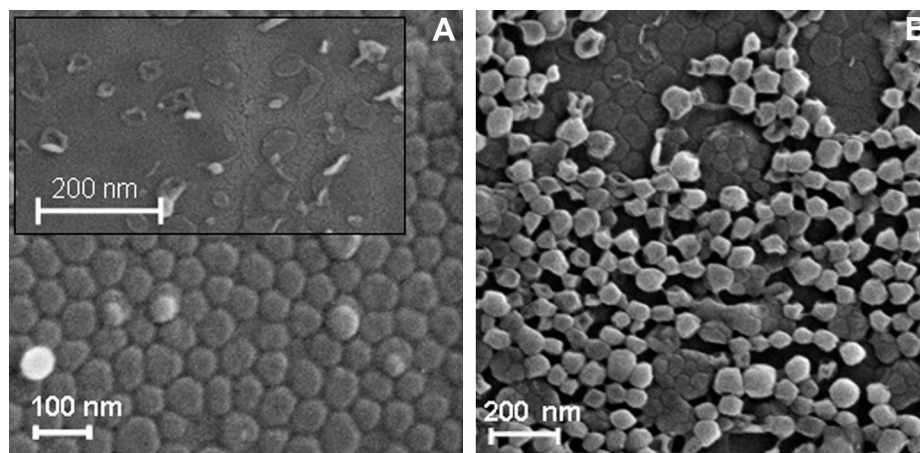


Fig. 2. Cryo-SEM images of (A) DPPC and (B) DSPC liposomes on mica. Inset in (A): DPPC-SUVs that have undergone fusion or rupture.

due to their rupturing upon adsorption to the mica surface, compounded by damage as a result of the cryo-SEM preparation procedure. DPPC liposomes were successfully imaged several times (Fig. 2(A)), but not always (Fig. 2(A) inset), possibly because they may rupture following exposure to the air–water interface during the preparation, as demonstrated by the AFM results, or because they are only marginally-stable to the cryo-SEM preparation procedure itself. The greater apparent stability of the DPPC-SUVs to rupture following passage through the air–water interface indicated by the cryo-SEM images (Fig. 2(A)), relative to that in the AFM (Fig. 1(C)), may be due to the fact that the water film coating the small cryo-SEM mica samples (see Experimental) was removed more gently. DSPC liposomes could always be imaged intact via cryo-SEM, and a typical micrograph is shown in Fig. 2(B). A loose, somewhat irregular liposome array – probably due to insufficient rinsing – is observed on top of the first, quite flattened liposome layer in contact with the mica substrate. We remark on the similarity of the DSPC-SUV cryo-SEM images to those taken for HSPC-SUVs on mica [25]. Generally DPPC and DSPC liposomes look similar upon adsorption to mica.

3.2. Normal interactions between mica surfaces following liposome adsorption

Normal forces between the mica surfaces across conductivity water following incubation in the liposome dispersions and rinsing were measured for the DMPC-, DPPC- and DSPC-SUVs and are shown in Figs. 3–5. Force axes are normalized as $F_n(D)/R$ with respect to the radius of curvature R of the mica surfaces so that different experiments can be compared within the Derjaguin approximation [36].

3.2.1. DMPC

Fig. 3 shows normal forces $F_n(D)$ as a function of separation D between two DMPC-SUV covered mica surfaces. Liposomes were adsorbed overnight at 20 °C and the force profiles determined at that temperature, to ensure that the surface-attached vesicles were fully in the SO phase (black filled and empty symbols) as is considered later. Measurements were also carried out at 28 °C (gray half filled and crossed symbols), either by heating the temperature-stable room to 28 °C for 2 h and determining $F_n(D)$ at the higher temperature, or in a different experiment (inset) where both incubation and measurements were at 28 °C. It can be seen in Fig. 3 that whichever way the experiments were carried out, temperature had little effect on the normal forces, as all profiles scattered within

the same range. As the surfaces approach, repulsion forces commence at separations in the range 40–160 nm, attributed largely to steric effects due to loose liposome overlayers on the surface-attached bilayers (the AFM micrographs for the DMPC do not show such overlayers as they may have been removed by the scanning tip, as seen for example in Ref. [25]) (see Fig. 1(A)).

On closer approach, a sharp increase in repulsion is observed (from about 10 to 12 nm) attributed to steric interactions between the opposing DMPC bilayers. No trend was observed for second and subsequent approaches compared to first approaches. Generally all approaches at the same contact position are similar. The final separations achieved were 4.8 ± 1 nm or 8.5 ± 1.2 nm, and may be attributed to either one or two compressed bilayers (the thickness of one DMPC bilayer was reported [34] to be 3.5 ± 0.1 nm). A final

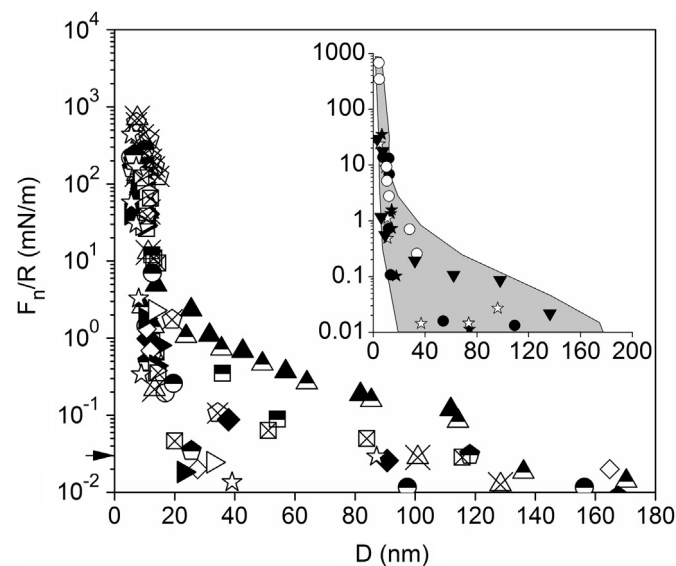


Fig. 3. Force $F_n(D)$ versus distance D profiles between DMPC-SUV coated mica surfaces across conductivity water normalized as $F_n(D)/R$ in the Derjaguin approximation, where R is the mean radius of curvature of the surfaces. The zero separation is defined with respect to air calibration. Filled symbols (first approach) and empty symbols (corresponding second approach) represent measurements performed at 20 °C, half filled symbols (first approach) and crossed symbols (corresponding second approach) are measurements performed at 28 °C after adsorption at 20 °C. Inset: measurements performed at 28 °C after adsorption at 28 °C. Gray area marks the range of force profiles of the main figure. Different shaped symbols correspond to different contact positions. The results shown are from three independent experiments. Arrow on y axis indicates the approximate noise level.

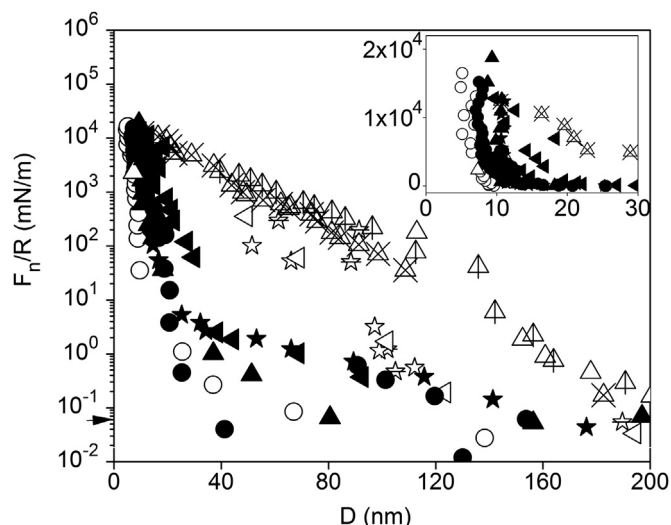


Fig. 4. Normalized $F_n(D)/R$ versus distance D profiles between DPPC coated mica surfaces across conductivity water. Only 'better' contact points are shown here (see text). Filled symbols are first approaches, empty symbols are second approaches and crossed symbols correspond to force profiles taken on separation of the surfaces. Open circles correspond to the only second approach for which the friction force remained low under high pressures. The final separation between the surfaces for all profiles is 8.8 ± 2 nm. Arrow on force axis indicates the approximate noise level.

separation of one bilayer is most likely a consequence of hemifusion-like rearrangement of areas with monolayer defects in the two opposing bilayers, as explained in the characterization section. It has also previously been described that fusion occurs as a result of mutual membrane proximity and exposure of hydrophobic regions [37]. For some contact points, small jumps-in were observed following strong compression.

From preliminary measurements of jumps-out following strong compression of the surfaces, we could estimate from the JKR relation [29] an interfacial energy which was comparable to that (≈ 15 mJ/m²) reported in Ref. [25]. Similar values were also reported for DPPE-DLPE bilayers [30] and attributed to hemifusion of monolayer deep holes.

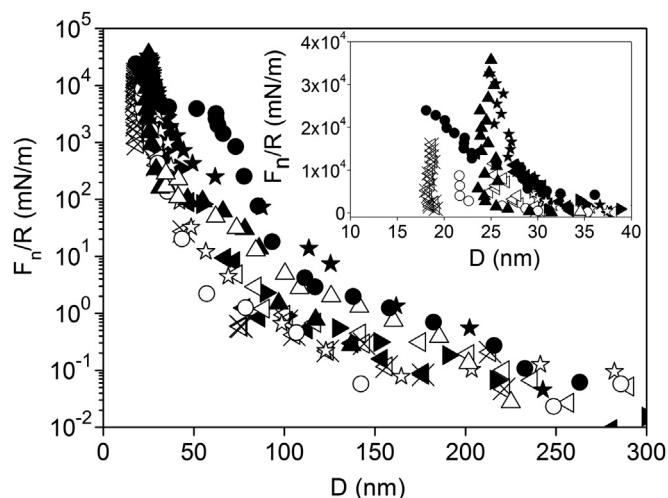


Fig. 5. Normalized $F_n(D)/R$ versus distance D profiles between DSPC-coated mica surfaces across conductivity water. Filled symbols are first approaches, empty symbols are second approaches and crossed symbols correspond to profiles taken on separation of the surfaces. Hard wall separations are (see inset) of 20 ± 3 nm and 27 ± 3 nm, corresponding to 2 or 3 confined liposome layers (see text).

3.2.2. DPPC

Normal interactions following incubation in DPPC-SUV dispersions revealed two distinct behaviors correlated with the lubrication ability of the surface-attached layers. Some 20% of the contact points, henceforth designated the 'better' ones, exhibited very good lubrication properties up to high pressures on a first approach of the surfaces, while other contact points exhibited considerably less good lubrication, which could be measured in the SFB only up to considerably lower pressures (frictional forces are described in Section 3.3). In the latter 'less good' cases, at higher pressures the frictional force exceeded the maximal shear force that could be applied and no sliding was observed between the surfaces as the top one was made to move laterally ('rigid coupling' behavior). Normal force profiles on first approaches for both types of contact points are similar. Fig. 4 shows force profiles only for these 'better' contact points, where the highest pressures reached correspond to the point at which rigid coupling was observed on shear. We note that profiles on separation of the surfaces following a first approach, as well as profiles on second and subsequent approaches at these 'better' contact points, show repulsion at much longer range compared to first approach profiles. This is a clear indication that the DPPC-SUV covered surfaces have been damaged following the first approach and shear, and is considered in the Discussion. The final 'hard-wall' separation at the highest compressions reached in our measurements was $D_f = 8.8 \pm 2$ nm for the 'better' contact points, corresponding to a separation of two bilayers, and was in the range 20 ± 4 nm for the 'less good' contact points, which might correspond to several stacked bilayers, or a less ordered configuration.

3.2.3. DSPC

Force profiles of DSPC liposome-covered mica surfaces are shown in Fig. 5, and exhibit two regimes similar to HSPC-SUV coated mica surfaces [25], which the DSPC-SUV indeed closely resemble. $F_n(D)$ increases roughly exponentially with decreasing D at large separations, and more sharply on closer approach where steric interactions between opposing liposome layers presumably set on (Fig. 5). The profiles show that first and subsequent approach profiles are similar to each other as well as to separation-profiles, indicating that the forces are both reversible and reproducible. Generally, second approaches are more inward compared to first approaches, which we attribute to squeeze-out of the overlayer of loosely bound SUVs from between the surfaces on the first compression, in line with similar observations on HSPC-SUV coated surfaces [25]. (These excess liposomes can be clearly seen in the cryo SEM and AFM images in Figs. 1 and 2). Upon compression, pressures of up to 180 atm could be reached without rigid coupling of the surfaces, with hard wall separations of 27 ± 3 nm and 20 ± 3 nm, roughly corresponding respectively to 3 and 2 confined liposome layers (the thickness of one bilayer is 5 nm, in line with ref [34], so that each fully-flattened SUV is ca. 10 nm in thickness).

3.3. Shear force measurements between mica surfaces following liposome adsorption

3.3.1. Shear force profiles

In some correspondence to their different surface structures and $F_n(D)$ profiles, the different liposomes also exhibited very different response to shear, as is shown in the shear-trace profiles in Figs. 6 and 7. Mica surfaces that had been incubated in DMPC-SUV dispersions provided good lubrication only up to relatively low pressures (Fig. 6). Though the normal force profiles of all DMPC experiments were similar (within the scatter), some different trends in the shear behavior were observed. For the experiments where incubation was at 20 °C and measurements at either 20 °C or 28 °C, the layers could withstand pressures up to ca. 25 atm, at

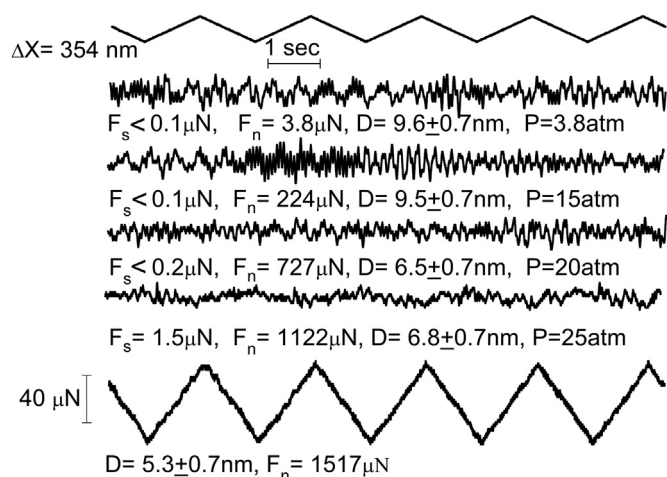


Fig. 6. Typical shear force versus time traces for DMPC system taken directly from the SFB at 20 °C. The upper trace shows the applied back and forth lateral motion of the upper surface Δx_0 – i.e. the actual lateral motion applied to the upper surface – while the other traces show the forces transmitted to the lower surface at different mean compressions P , corresponding separations D and friction and normal forces, F_s and F_n .

which point the friction between them increased abruptly and the surfaces became rigidly coupled (up to the maximal lateral motion applied to the top surface). At lower pressures, however, the shear forces were very low (Fig. 6). For the case where both incubation and measurements were carried out at 28 °C, already at pressures of some 12 atm the frictional forces became abruptly larger than the maximal applied shear forces in our system, resulting in rigid coupling. It should also be noted that once pressures were reached at which the friction increased abruptly, the friction forces observed at the same contact point on subsequent approaches were always higher for a given load, and set on at larger surface separations, indicating damage/disruption of the surface layers. We note, importantly, that the abrupt onset of high friction and the rigid coupling regime mentioned above came at the same time as adhesion between the surfaces, with a corresponding jump-out on separation, as previously described for the DMPC case.

Mica surfaces that had been incubated in DPPC-SUV dispersions exhibited better lubricating properties compared to those incubated in DMPC-SUV, though generally much less efficient than the

lubrication exhibited by HSPC-SUV coated surfaces [25]. For about 80% of the contact points examined, referred in the text as ‘less good’ contact points, high friction and consequent rigid coupling on first approach of the surfaces was reached at pressures of 10–20 atm on average. Only at about 20% of the contact points, designated as ‘better’ ones, was the lubrication very efficient and the frictional forces low on the first approach of the surfaces, up to the highest pressures reached in our study (130 atm) (Fig. 7(A); other ‘better’ contact points showed similarly low friction up to 130 atm). However, this very good lubrication was not reproducible on second and subsequent approaches, where high friction with rigid coupling was indicated at just a few atmospheres pressure at much larger surface separations, in line with the normal force profiles (Fig. 4). The indications therefore were that high friction at these ‘better’ contact points (on second and subsequent approaches) was correlated with damage of the DPPC-SUV layers, manifested also as much thicker surface layers.

For mica surfaces that had been incubated in DSPC-SUV dispersions, very good lubrication was achieved at all contact points and all approaches up to the highest pressures (~180 atm) measured, both on first and subsequent approaches, with no evidence of surface damage. Typical shear traces are shown in Fig. 7(B).

3.3.2. Friction coefficients

The sliding friction coefficient may be evaluated as the derivative of the shear (friction) force during sliding as a function of the normal load; an ‘effective’ friction coefficient is evaluated simply by dividing the friction by the load (the two are equivalent if the friction coefficient is load-independent). Friction versus normal forces are summarized in Fig. 8(A) for the DPPC-SUV incubated mica surfaces, for first approaches in the ‘better’ contact points; and in Fig. 8(B) for the DSPC-SUV incubated mica surfaces, with the corresponding friction coefficients indicated for the different profiles. The range of loads and frictional forces measured for the DMPC-coated surfaces is shown as a shaded band in Fig. 8(A). For the DPPC-SUV coated surfaces the friction coefficients were ca. 1.8×10^{-4} on average, up to the highest pressures (130 atm) reached in our study (prior to the ‘rigid coupling’ regime beyond which sliding could not be induced), though always with the caveat that this was for the 20% or so of ‘better’ contact points, and only on first approach. For the DSPC-SUV coated surfaces the results were reproducible and reversible at essentially all points measured, with

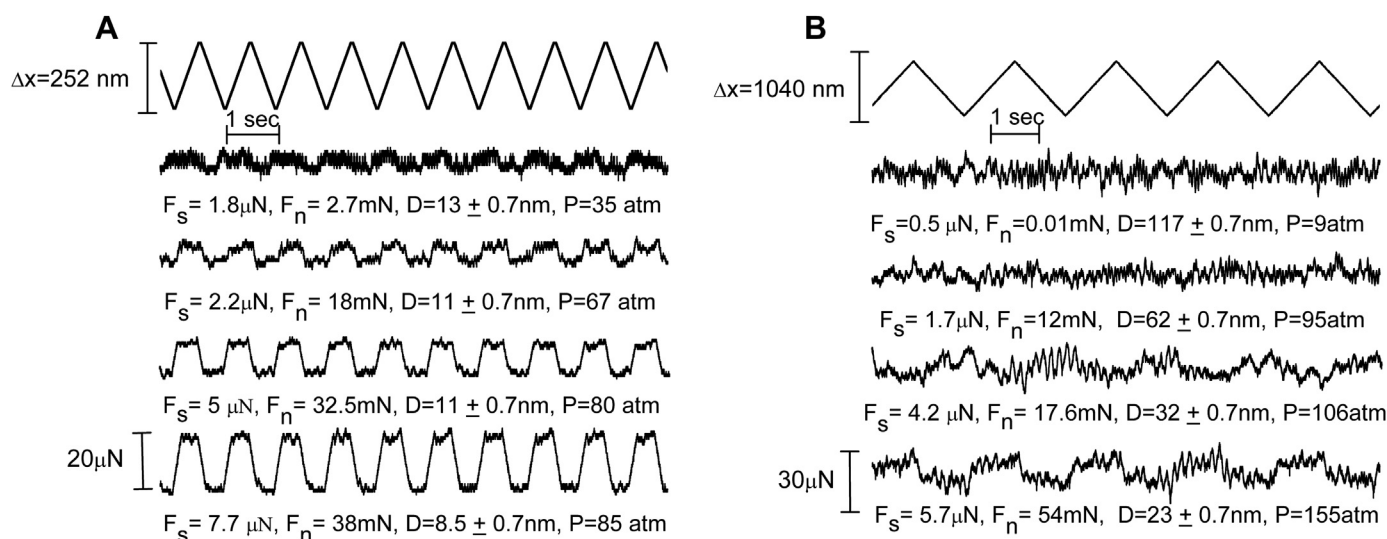


Fig. 7. Typical shear force versus time traces for the (A) DPPC-SUV coated and (B) DSPC-SUV coated mica surfaces. Pressures were calculated using Eq. (2) based on $K = 5 \times 10^9$ N/m². When calculating P based on maximal flattening, the values are (from top down): (A) 48,90,110,115 atm and (B) 8,82,92,134 atm.

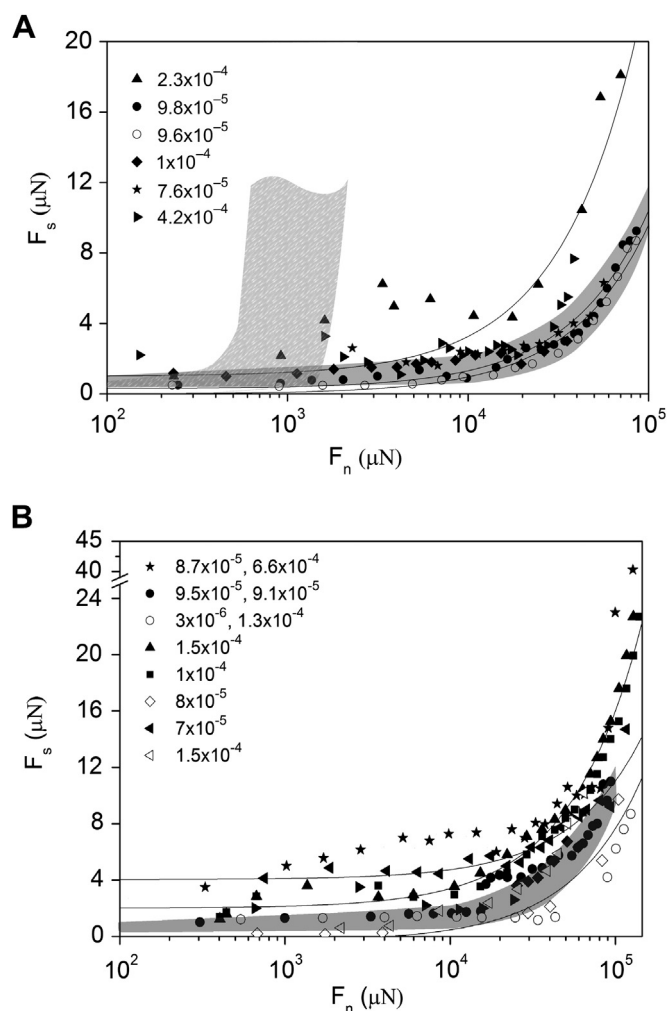


Fig. 8. Friction forces F_s as a function of applied loads F_n between (A) two DPPC-SUV coated mica surfaces on a first approach (for the ca. 20% of 'better' contact points), and (B) between two DPPC-SUV coated mica surfaces. Full symbols- first approaches, empty symbols- second approaches at the same contact position. Friction coefficients are estimated by fitting each data set to the relation $F_s = \mu F_n$. Several fitting curves are shown. Friction coefficients for the shown data series appear in the insets. The broad textured gray band in Fig. 9A indicates the range of DMPC friction data. The shaded band in Fig. 9B corresponds to the narrow band in Fig. 9A, which includes the bulk of the data points for the DPPC-SUV coated surfaces.

some scatter as seen in Fig. 8(B), and also at subsequent approaches at the same contact position.

3.3.3. Effect of shear velocity

The dependence of the sliding friction on the sliding velocity was examined and is shown in Fig. 9 for both DPPC-SUV- and DSPC-SUV-coated mica surfaces, at two different compressions for each lipid (for the DPPC-SUV the data is for the 'better' contact points on a first approach). It can be seen that there is little variation, within the scatter, over some two orders of magnitude in sliding velocities.

3.3.4. Prolonged sliding times

In order to determine the stability of the liposomes as boundary lubricants, we have also examined the lubrication properties of the PC-SUV-coated surfaces at longer sliding distances and sliding times. As each shear profile, as shown for example in Figs. 6 and 7, usually takes less than 30 s, prolonged shear was applied for 1 h at two different compressions during several profiles, both with DPPC-SUV coated- (on first approach at the 'better' contact points)

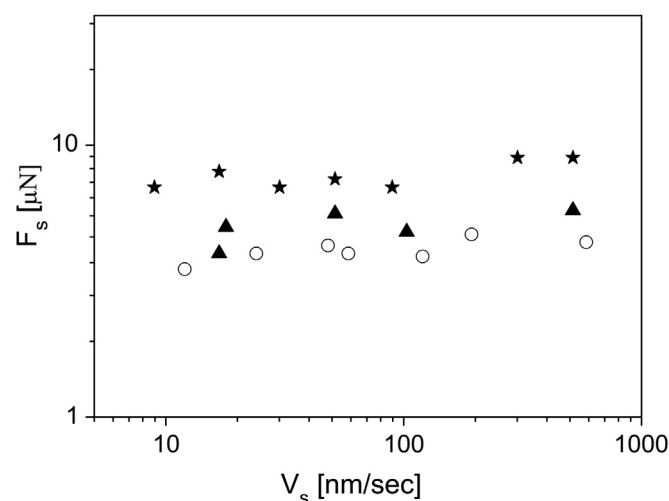


Fig. 9. Friction force variation with sliding velocity between DSPC-SUV-coated (filled symbols) and DPPC-SUV coated surfaces (empty symbols, for the 'better' contact points on a first approach). ★-20 nm, 180 atm, ▲-29 nm, 140 atm, ○-10 nm, 60 atm.

and DSPC-coated surfaces. It was discovered that the frictional forces not only remain very low, but actually decrease slightly (roughly by a half) as a result of prolonged shear under pressure for both types of lipids, as seen in Fig. 10 for the DSPC-SUV coated surfaces (DPPC-SUVs not shown).

4. Discussion

The main findings of this study concern the lubrication efficiency of surface-attached SUVs of the homologous series of di-acyl phosphatidylcholines, DMPC (C_{14}), DPPC (C_{16}) and DSPC (C_{18}). The results show clearly that at the highest pressures – up to 100 atm or higher – the DSPC-SUV layers are capable of providing very efficient lubrication, while the DPPC-SUV can also do this, though on a first approach only, and only at a fraction of the contact points between the surfaces. The low friction is attributed, as for the HSPC-SUV layers [24,25], to the hydration lubrication effect afforded by the exposed phosphocholine groups, which are known to be highly hydrated. In this mechanism the hydration water molecules about these headgroups are strongly attached, thus withstanding high normal stresses without being squeezed out, while their rapid relaxation ensures a fluid-like response on shear, thereby leading to very low friction coefficients. This effect is not specific to

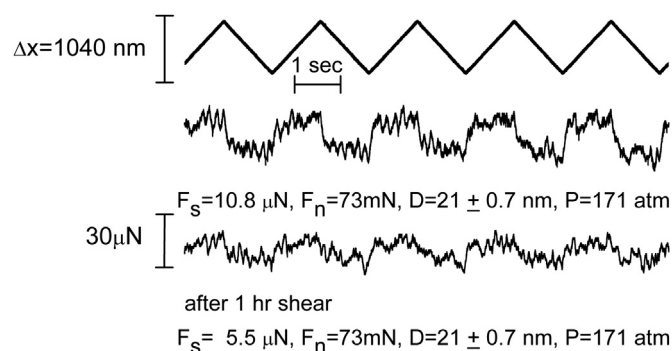


Fig. 10. Typical shear force versus time traces for the DSPC system taken directly from the SFB for one contact position. The top trace shows the applied back and forth lateral motion of the upper surface Δx_0 . The middle trace shows the response motion at the indicated compression, while the bottom trace shows the response motion at the same compression after applying continuous shear motion for 1 h. Corresponding friction forces F_s are also shown. The shear force after 1 h is reduced by half.

phosphocholine groups and has been demonstrated with several other hydrated species [38–40] as well as between bare mica surfaces sheared in salt solution [21,41]; boundary lubricants based on the phosphocholine groups [25,40] appear to provide lubrication up to the highest pressures, presumably due to their being most strongly hydrated [42–48]. For both DSPC and DPPC the temperature of measurements (ca. 25 °C) was well below the respective main FSB transition temperatures of 54 °C and 41 °C of these PCs, and, as we shall see, even further below the transition temperatures of the respective *supported* bilayers [49,50]. Thus both these surface-supported SUVs were well within their gel phase.

The DMPC layers, in contrast, are not robust at these high pressures. Here it is important to note that the SO-to-LD transition for supported DMPC bilayers on mica has been studied in detail by Charrier and Thibaudau [49]. They find two SO-to-LD transition temperatures, which they identify with the lower leaflet of the bilayer (in contact with the mica) and the upper leaflet (in contact with the aqueous phase), respectively. Both of these transition temperatures are significantly higher than the main FSB transition temperature for DMPC, with the transition temperature of the lower leaflet being in turn higher than that associated with the upper leaflet. Since in our experiments we deal with supported liposomes or bilayers attached to mica, the relevance of these results to our own study is clear: The effect of attachment to the (mica) substrate is to increase the stability of the leaflets adjacent to it, so that a higher temperature is required to induce an SO-to-LD transition, and we expect this to apply in all three PC systems studied. It is found [49] that the upper leaflet of the mica-attached DMPC bilayer – which is the one at the slip-plane as the DMPC-bilayer-coated surfaces slide past each other in our study – is fully in the SO phase up to 25 ± 1 °C, when the transition to the LD phase commences, and is fully in the LD phase by 33 ± 0.5 °C (for the lower DMPC leaflet the SO to LD transition takes place over the temperature range 38.5 °C–44.5 °C). Thus at the temperatures at which DMPC measurements were made, 20 °C and 28 °C, the upper leaflet of the bilayers was, respectively, fully in the gel phase or in a mixed SO/LD phase. One might expect similar increases in the transition temperatures (of both upper and lower leaflets) to be relevant also for the DPPC and the DSPC bilayers in contact with the mica. For the case of intact liposomes sliding past each other (as for some of the DPPC and all of the DSPC measurements), however, the interacting surfaces are those of the outer leaflets of the *second* bilayer away from the mica surface. For such leaflets – separated by 3 monolayers from the solid surface – one would expect a much smaller effect of the supporting substrate on the transition temperature, and that their SO to LD behavior be much closer to that of the FSBs of the respective phospholipids. The effect of pressure in these experiments on the transition temperatures [51] is small, and may be shown to increase the LD-to-SO transition by an amount in the range ca. 0.7 °C–5 °C for the data presented in this study.

As seen clearly by AFM (Fig. 1(A)), the DMPC-SUVs (di-C₁₄ PCs) ruptured spontaneously on adsorption to mica (AFM micrographs were taken following overnight adsorption). The AFM measurements were at room temperature (25 °C, where the bilayers are still essentially fully in their SO phase). That, together with the similarity of the normal force profiles and of the frictional behavior for the DMPC-coated surfaces at both 20 °C (SO phase) and 28 °C (mixed SO/LD phase), suggests that such rupturing occurs at both temperatures prior to the SFB measurements. There are adhesive forces acting between the mica and the SUV membranes, as revealed by the measured surface energy evaluated from the pull-off [25], which likely arise from the interactions between the dipole on the phosphocholine groups and the charges on the mica surface. These adhesive forces act to rupture the adsorbed SUV bilayer membrane through tension at the liposome/mica

bifurcation line and are opposed by interactions tending to keep the SUV membranes intact, which arise from the hydrophobic forces between the PC tails in the bilayer. Since the opened-out configuration of the vesicles into a lamellar bilayer phase is also the thermodynamically-favored one, the closed liposome structure on the surfaces is therefore metastable. Presumably for the shorter acyl chains (C₁₄) of the DMPC-SUVs the adhesive forces to the mica rapidly overcome the chain–chain hydrophobic attractions, so that rupture occurs spontaneously on adsorption at short times (see below). Corresponding with this was a relatively poor lubrication behavior (Fig. 6 and light-shaded band in Fig. 8(A)) at higher pressures: As the increase in the friction occurs abruptly (see bottom two traces of Fig. 6) we may attribute it to the onset of hemifusion between the DMPC bilayers, possibly facilitated by holes in the layers, with some squeeze-out of lipids from between the surfaces. It is for this reason that the DMPC-SUVs fail to be robust, and thus efficient, boundary lubricants at higher pressures.

For the case of DSPC-SUVs (di-C₁₈ PCs), in contrast, the vesicles adsorbed and retained their closed shape even when taken through an air/water interface (Figs. 1(D) and 2(B)), and likely even when compressed to high pressures and sheared (as revealed by separate experiments, R. Sorkin et al., to be published). This is in line also with the robust behavior of the HSPC-SUVs [24,25], where retention of the closed nature of the vesicles was also indicated under compression and shear, and is attributed to the much larger energies (relative to DMPC) required to rupture the DSPC bilayers due to the 4 additional –CH₂– groups on each PC tail. This effect expresses itself also in the much higher SO-to-LD transition temperature for these supported liposomes, more than 30 °C higher than the room temperature of the measurements. Correspondingly, DSPC-SUVs also exhibited extremely efficient lubrication properties arising from the hydration lubrication mechanism (Figs. 7 and 8), similar to those previously reported for HSPC liposomes [24,25], which indeed they resemble in their composition. Thus although the DSPC-SUVs are in a metastable state when adsorbed on the mica (as the thermodynamically-stable configuration is that of a continuous bilayer on the mica), they are much longer lived than the adsorbed DMPC-SUVs.

We may make a rough estimate of the relative stabilities of the liposomes on the surface as follows. For each two additional –CH₂– groups on a PC tail there is an additional interaction energy $\delta\epsilon = \alpha k_B T$ say, required to disrupt the liposome. This arises from hydrophobic interactions, modulated by some difference in hydration energies of the neighboring hydrated phosphocholine headgroups (since the longer tailed bilayers will have a higher areal density of headgroups [52]). Thus one requires roughly $\Delta E \approx 4\delta\epsilon \approx 4\alpha k_B T$ additional energy to disrupt the DSPC-SUVs relative to the DMPC-SUVs bilayers. The magnitude of $\delta\epsilon$ for the disruption process is not precisely known, but may be estimated from the properties of the liposomes, in particular data on the relative critical micelle concentrations of liposomes with different acyl chain lengths, which would suggest $\alpha \approx 2$ [53]. Thus $\Delta E \approx 8k_B T$. If we assume a simple Eyring transition-state model [54] to describe the rupture process, with mean time $\tau_{\text{rupture}}(\text{PC})$ for rupture of an SUV bilayer, then we might expect $\tau_{\text{rupture}}(\text{DSPC}) \approx \tau_{\text{rupture}}(\text{DMPC}) \times e^{\Delta E/k_B T} \approx (3 \times 10^3) \tau_{\text{rupture}}(\text{DMPC})$. Thus even if the DMPC-SUV rupture within minutes of their adsorption on the mica [50,55], we might expect the adsorbed DSPC-SUVs on the surface to be stable for up to days or longer, which is consistent with our observations.

The DMPC-SUVs (di-C₁₄ PCs) and the DSPC-SUVs (di-C₁₈ PCs) thus represent two extremes in this study: one might expect the surface-attached DPPC-SUVs (di-C₁₆ PCs) to be intermediate between these two cases. And indeed the behavior of the DPPC, both in terms of the surface structure and the response to shear, was intermediate

between that of easily-ruptured DMPC-SUVs and that of very-stable DSPC-SUVs. Thus while the DPPC-SUV adsorbed onto mica retaining their integrity and closed structure when they remained immersed in water throughout, as shown by AFM (Fig. 1(B)), the additional stress on the liposomes due to passage through the air/water interface could cause them to rupture into bilayers (Fig. 1(C)). This suggests that their resistance to rupture when adsorbed on the mica surfaces was marginal. Nonetheless, as seen by cryo-SEM (Fig. 2(A)) the intact liposomes could survive the cryo-SEM preparation procedure – which also involved removing a water film from the liposome-coated mica. Moreover, AFM micrographs (not shown) of the lens-mounted mica surfaces following removal from the water (en route to mounting in the SFB) also revealed regions of the surfaces which were covered by intact liposome layers. We believe these regions correspond to those parts of the mica which remained wetted by patches of water on transferring from the rinsing beaker to the water-filled SFB bath (see Surface Preparation Procedure in Experimental Section), and so did not experience the shear stress associated with crossing the water/air interface.

The DPPC case is of special interest as this phospholipid is ubiquitous in biological environments, including synovial joints where lubrication is at a premium [56]. Under compression and shear, the frictional response (or lubricating ability) of the DPPC-SUV layers fell into one of two categories. In a minority of the cases (ca. 20%, termed the ‘better’ contact points) the lubricating ability on a first approach was very good, with mean values of $\mu \approx 1.8 \times 10^{-4}$ up to $P = 130$ atm (Fig. 8(A)), at a ‘hard-wall’ separation of ca. 9 nm, corresponding to 2 trapped bilayers (Fig. 4). We attribute this to the presence of intact DPPC-SUV layers at these contact points: the low fraction of these ‘better’ contact points is due to the relatively small areas of the mica substrate on which intact liposome layers survived the passage through the water interface through remaining in a wetted patch, as described above. The boundary lubrication behavior of these DPPC-SUV coated regions may be attributed as follows: On compression and shear, the DPPC-SUVs confined in the contact region ruptured and fused into the lamellar phase, initially comprising 4 bilayers (corresponding to two flattened liposome layers). The two central bilayers, no longer a part of any liposomes, were then squeezed out, piling up as ‘debris’ at the edges of the contact zone, as illustrated schematically in Fig. 11. The two remaining bilayers – one attached to each surface – then slid in smooth contact past each other, with little additional material confined between them, with the very low friction arising from the hydration lubrication mode. On separation from the initial approach and compression, it appears that the irregular ‘debris’ at the edges of the contact zone is ‘sucked’ or diffuses back into the contact region between the surface-attached bilayers. Thus on second and subsequent approaches the onset of steric repulsion at these initially-‘better’ contact points is generally much further ranged (see Fig. 4), and on compression the material separating the surface-attached bilayers is likely irregular and not easily squeezed out (see schematic in Fig. 11). At the same time, the sliding friction across such confined irregular material – in the second and subsequent approaches – is likely to be greater (as is indeed observed) than between two bilayers alone on first compression and sliding, due to energy dissipation arising from the shear and deformation of the trapped ‘debris’.

In contrast to the above scenario, in the case of most of the contact points examined between DPPC-SUV-bearing mica surfaces (some 80%, termed ‘non-lubricating’), the lubrication was relatively poor above ca. 10–20 atm mean contact pressure. These ‘less-good’ contact points correspond, we believe, to the regions that emerged dry on withdrawal through the air–water interface (see Surface Preparation Procedure in Experimental Section), and thus experienced stronger lateral stresses, which led to the SUV

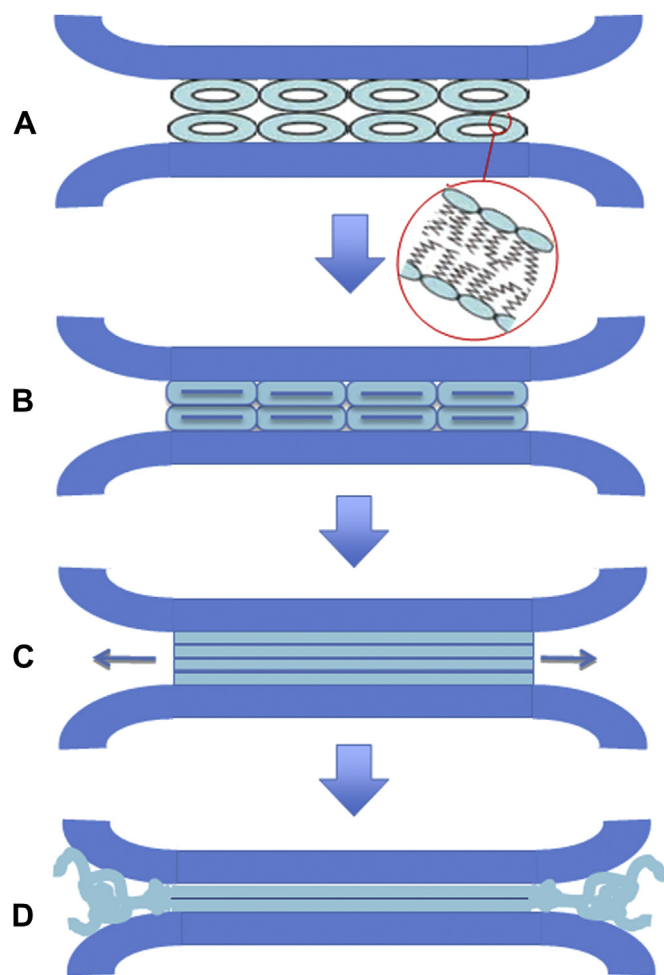


Fig. 11. Schematic representation of the stages of structural change DPPC SUVs undergo upon compression and shear. A Two opposing surfaces are covered with SUVs. B Upon compression, the vesicles are flattened and most of their water content is squeezed out. C The edges of vesicles merge so that a stack of bilayers is formed. D Upon shear two middle bilayers that are not directly attached to mica are squeezed out from between the surfaces. The excess squeezed out material results in onset of steric repulsion at higher separations upon subsequent approaches.

rupture and to debris formation, as seen in the inset to Fig. 2(A). The ‘hard-wall’ on strong compression for these contacts was in the range ca. 15–25 nm. This suggests that both normal and shear interactions at these ‘less good’ contact points involved the trapping between the surfaces of irregular material that could not readily be squeezed out from between the two surface-attached bilayers. Repulsive steric forces therefore set on at larger separations, while the associated sliding friction across such confined material will be greater than between two bilayers alone, due to energy dissipation arising from the shear and deformation of the trapped irregular material.

There is in addition one suggestive observation that may be made concerning the relative friction forces between DSPC-SUV layers and those due to the DPPC layers (at the ‘better’ contact points) described above. While there is significant scatter in the variation of F_s with F_n for both the DPPC-SUV and the DSPC-SUV-coated surfaces (a possible reason for this is discussed below), examination of Fig. 8 reveals that the bulk of the data for the DPPC case, (dark shaded band in Fig. 8(A)), corresponds to lower friction forces than the bulk of the data for the DSPC case, as clearly seen by superposing the shaded band in Fig. 8(B). The reason for this may be that for the DPPC case the sliding surfaces are in the lamellar

bilayer configuration, while in the DSPC they are, we believe, in a closed, flattened vesicular configuration, exposing a slightly different topography. However, it may also be – and we believe this is closer to the correct explanation – due to the fact that the weaker interactions of the C₁₆ tails of the DPPC result in a laterally-less-dense bilayer [52]. One therefore expects for the DPPC a more highly-hydrated phosphocholine headgroup layer, than is the case for the DSPC bilayers, which are denser due to the stronger attraction of the C₁₈ tails. Since the efficiency of the hydration lubrication mechanism depends on the extent of hydration, this is consistent with the lower friction of the DPPC boundary lubricants relative to the DSPC ones, as seen in Fig. 8(B).

The very weak dependence of the sliding friction on the sliding velocity vs. Fig. 9, is consistent with the boundary lubrication regime, where the friction coefficient varies only very weakly with sliding speed [57].

Finally, we comment on the robustness of the PC-SUV layers under prolonged sliding friction, which is of importance both from the point of view of understanding their limiting properties, and also from a practical point of view. We found that both the DPPC lamellar phase (in the ‘better’ contact positions on the first approach) and the DSPC-SUV layers continued to provide extreme friction reduction under prolonged shear of at least 1 h, as seen in Fig. 10 (for DSPC; DPPC not shown). For both these PC’s, friction not only remained low but in fact decreased slightly (by a factor 2 or so following 1 h sliding), possibly due to rearrangement of the surface layers under shear to a more favorable orientation with respect to the sliding direction. The important point, however, is that the DSPC-SUVs show no sign of deterioration with respect to their very efficient lubrication ability even after 1 h of back-and-forth sliding at the same contact position.

5. Conclusions

A systematic study of the structural and friction-reducing properties of the series of DMPC, DPPC and DSPC liposomes adsorbed on surfaces was conducted, revealing several features concerning the nature and the determinants of boundary lubrication by PC vesicles. Overall, we found that the DSPC-SUVs were better boundary lubricants than DPPC-SUVs which were in turn better than the DMPC-SUVs, but this appeared to be due to two effects: robustness of the SUVs arising from the longer tails, on the one hand, and efficiency of the hydration lubrication mechanism on the other. Our results suggest that the earlier conjecture [24,25], that PC-SUVs are very efficient boundary lubricants in their SO phase but poor ones in the LD phase is too simplistic, and that the actual situation is more nuanced. Thus, we conclude that it is not sufficient for the surface-attached vesicles to be in their SO phase for efficient lubrication to occur. Rather, at a given temperature, there appears to be a progressive change in the response to compression and shear of surface-attached PC-SUVs, as they move deeper into the SO phase, corresponding to longer acyl tails – and thus stronger attractions between them – resulting in more robust liposomes. This, together with the greater hydration for PCs of lower transition temperatures (shorter di-acyl chains on the PCs) noted above, suggests that these factors may be combined for the optimal choice of PC-SUVs for high pressure boundary lubrication in aqueous media.

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Corrigendum

Corrigendum to “Origins of extreme boundary lubrication by phosphatidylcholine liposomes” [Biomaterials 34 (2013) 5465–75]

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The authors regret that a change is required to the acknowledgment section of this published manuscript.

The acknowledgment section of the above paper included the following statement:

“We acknowledge the contribution of Yechezkel Barenholz, Rivka Cohen and Hiba Kanaan of the Department of Biochemistry at the Hadassah Medical School of the Hebrew University, Jerusalem, who prepared and characterized some of the liposomes used in this study, and Y. Barenholz for suggestions and comments; we thank them for their contribution and regret they would not agree to be co-authors on this paper.”

Since publication of the paper, Professor Barenholz has registered a protest with the Editor-in-Chief about this acknowledgement. The acknowledgement section should therefore be altered to:

“We thank the European Research Council (Advanced Grant HydrationLube), the Charles McCutchen Foundation and the Israel Science Foundation for support of this work. We thank Ronit Goldberg, Jasmine Serore, Gilad Silbert and Guy Brand for useful discussions. Thanks also to Jacob Israelachvili and Adrian Parsegian for useful correspondence. JK particularly thanks Matthew Tirrell for useful advice. This research was made possible in part by the historic generosity of the Harold Perlman family.”

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